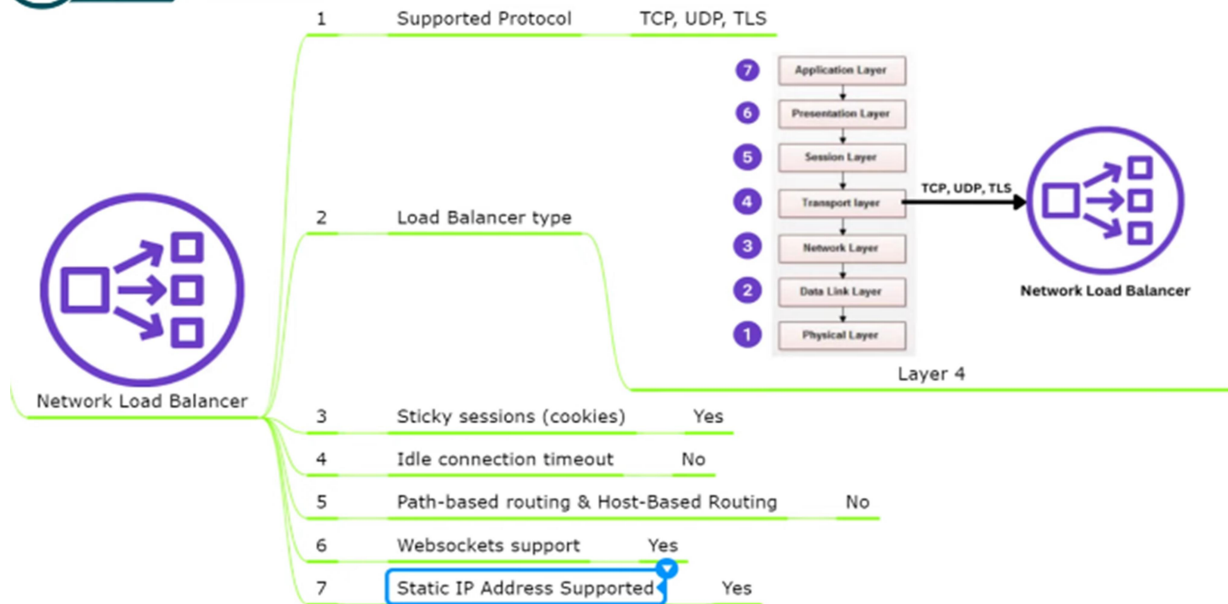
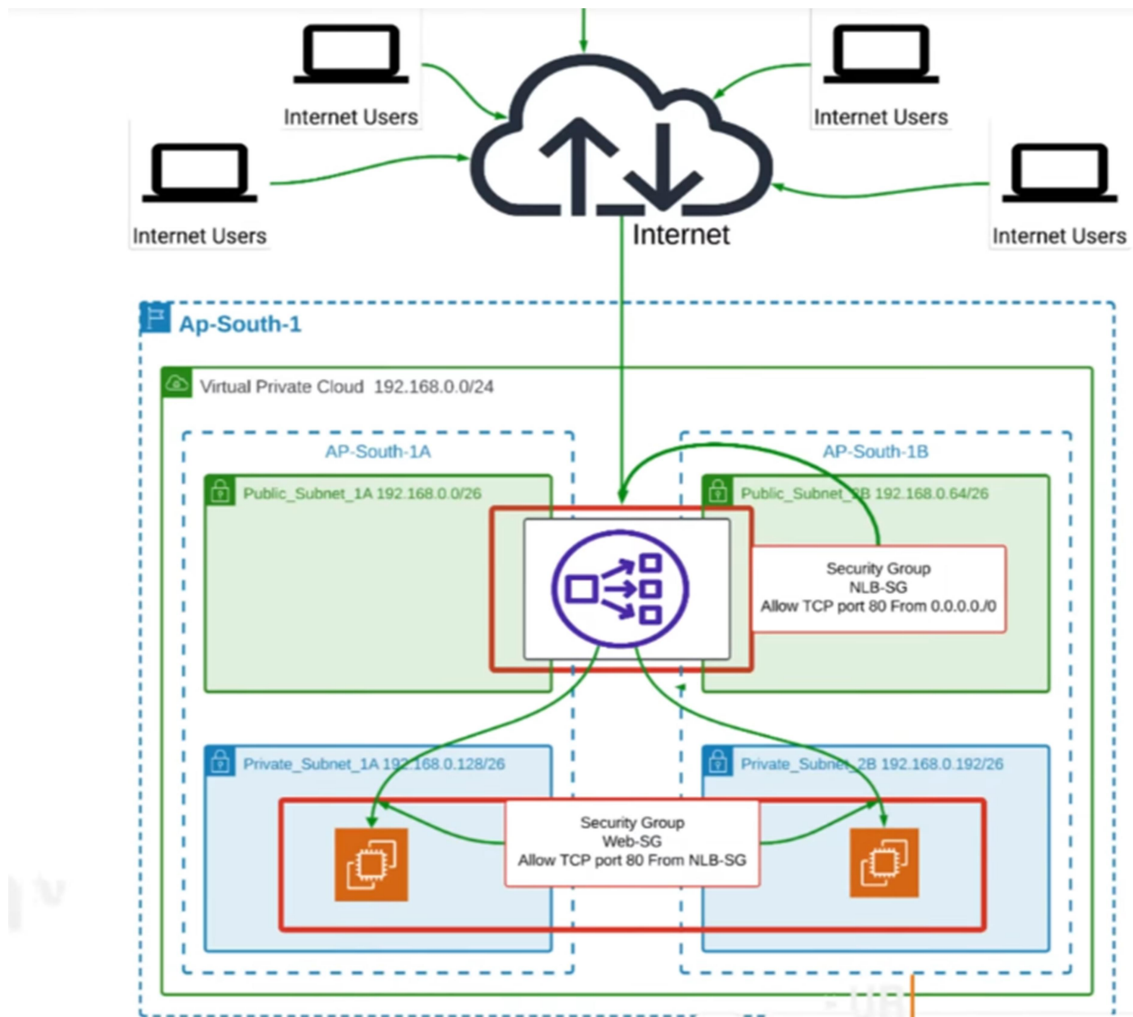


Network Load Balancer :

- 1 Network Load Balancers are best suited for load balancing of TCP traffic where extreme performance is required.
- 2 Operating at the connection level (Layer 4)
- 3 Network Load Balancers are capable of handling millions of requests per second, while maintaining ultra-low latencies.

Day 100





As per diagram create vpc and nlb-sg ,web-sg carefully.

Create instance as per diagram and userdata script to both server (web-server-1 and web-server-2)

```
#!/bin/bash
```

```
yum update -y
```

```
yum install httpd -y
```

```
systemctl start httpd
```

```
systemctl enable httpd
```

```
echo "Welcome to Web-Server-1" > /var/www/html/index.html
```

create Network Load-balancer

go to load balancer and create load balancer and select network load balancer

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

NLB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

An internet-facing load balancer routes requests from clients over the internet to targets. Requires a public subnet. [Learn more](#)

☐ Internal

An internal load balancer routes requests from clients to targets using private IP addresses.

IP address type [Info](#)

Select the type of IP addresses that your subnets use.

☒ IPv4

Recommended for internal load balancers.

☐ Dualstack

Includes IPv4 and IPv6 addresses

VPC

Select the virtual private cloud (VPC) for your targets or you can [create a new VPC](#). Only VPCs with an internet gateway are enabled for selection. The selected VPC can't be changed after the load balancer is created. To confirm the VPC for your targets, view your [target groups](#).

MyVPC

vpc-0d540412b8c0c8065
IPv4: 192.168.0.0/24

Mappings

Select at least one Availability Zone and one subnet for each zone. We recommend selecting at least two Availability Zones. The load balancer will route traffic only to targets in the selected Availability Zones. Zones that are not supported by the load balancer or VPC can't be selected. Subnets can be added, but not removed, once a load balancer is created.

☒ ap-south-1a (aps1-az1)

Subnet

subnet-00a0f7c2a42ac21b9

Public_Subnet_1A ▼

IPv4 address

Assigned by AWS ▼

☒ ap-south-1b (aps1-az3)

Subnet

subnet-0a8631cb987887057

Public_Subnet_2B ▼

IPv4 address

Assigned by AWS ▼

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security

Security groups - recommended

Security groups support on Network Load Balancers can only be enabled at creation by including at least one s for your load balancer must allow it to communicate with registered targets on both the listener port and the h are enforced on PrivateLink traffic; however, you can turn off inbound rule evaluation after creation within the l

Select up to 5 security groups

NLB-SG

sg-05ad5061d68adf78f VPC: vpc-0d540412b8c0c8065



Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener TCP:80

Remove

Protocol Port

TCP : 80

1-65535

Default action [Info](#)

Forward to Select a target group

Create target group

Listener tags - optional

Target type : instance choose

Target-1

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port

Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation

TCP 80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select vpc: myvpc and health checks as it is. Click next

Register targets

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you

Available instances (2/2)

Filter instances

☒	Instance ID	Name	State	Security
☒	i-02a38f8c9a8cd22c5	Web-Server-2	Running	Web-S
☒	i-02b1e4a9a9c19fcb6	Web-Server-1	Running	Web-S

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

click create target group

Protocol: TCP Port: 80

Default action: Forward to Target-1

Target type: Instance, IPv4

TCP

Create target group

create load balancer

Now test nlb dns and pasted to browser

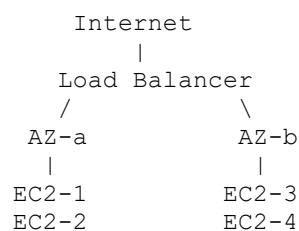
Note: based on client's same ip add it will divert traffic to single server unlike alb but it doesn't mean that server 2 is not healthy.

Cross zone load balancer #####333

Cross-Zone Load Balancing controls whether a load balancer can send traffic to targets in **all Availability Zones** or only to targets in the **same Availability Zone**

- **Cross-Zone Disabled**

Each load balancer node sends traffic **only to targets in its own AZ**.



If the LB node in **AZ-a** receives 100 requests:

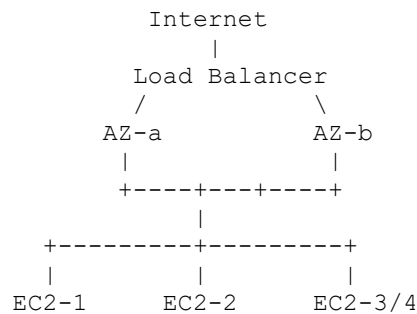
It does **not** send those requests to EC2-3 or EC2-4 in AZ-b.

Advantage: Traffic stays within the AZ, which can reduce cross-AZ data transfer.

Problem: If AZ-a has fewer targets than AZ-b, its targets may receive more traffic.

- **Cross-Zone Enabled**

Each load balancer node can distribute traffic across targets in **all enabled AZs**.



To Enable cross zone lb go to created nlb and select

